

EE 3202.304

ECE Fundamentals Lab II

Lab 6: MOSFETs and Digital Circuit Elements (CMOS)

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I. Experimental Results

Part 1 – Threshold Voltage

Threshold Voltage (n-channel)

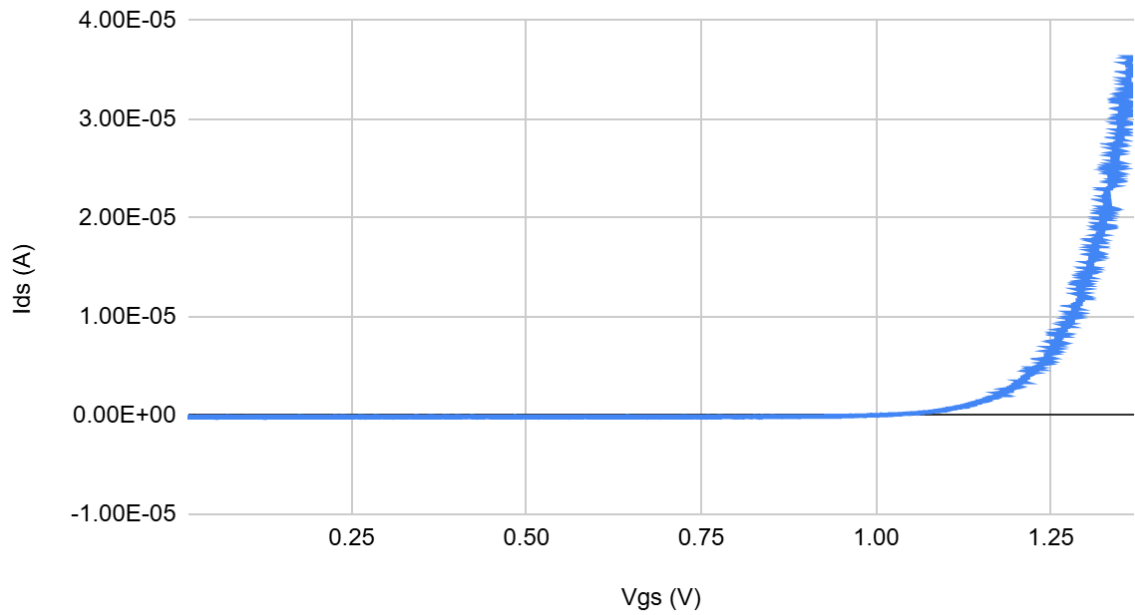


Figure 1.1: Threshold Voltage graph of n-channel

Threshold Voltage (p-channel)

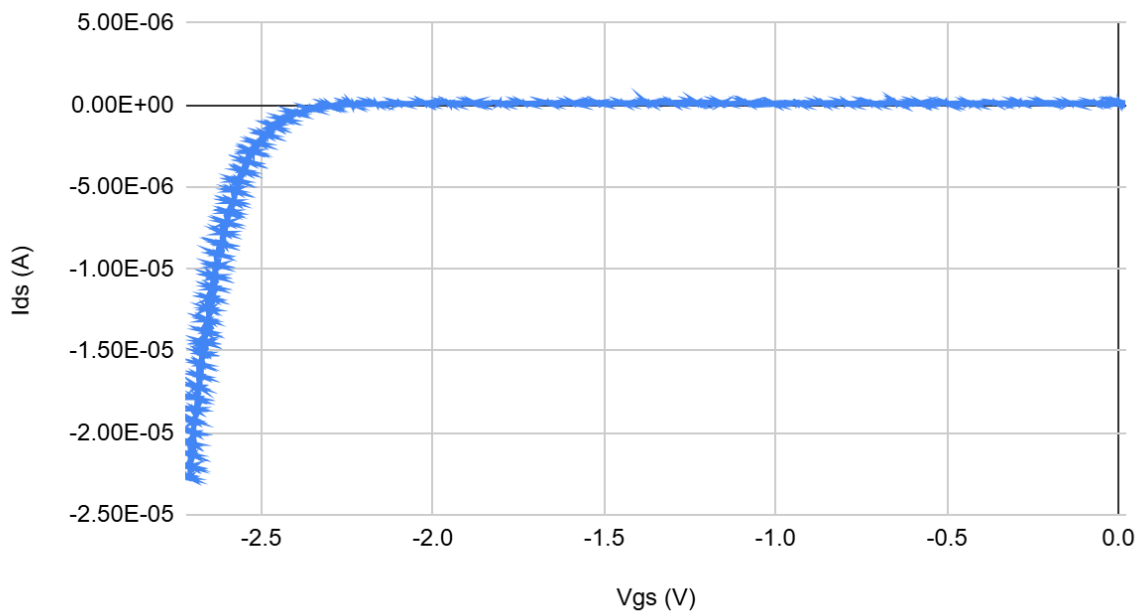


Figure 1.2: Threshold Voltage graph of p-channel

Part 2 – Curve Traces

Curve Trace (n-channel)

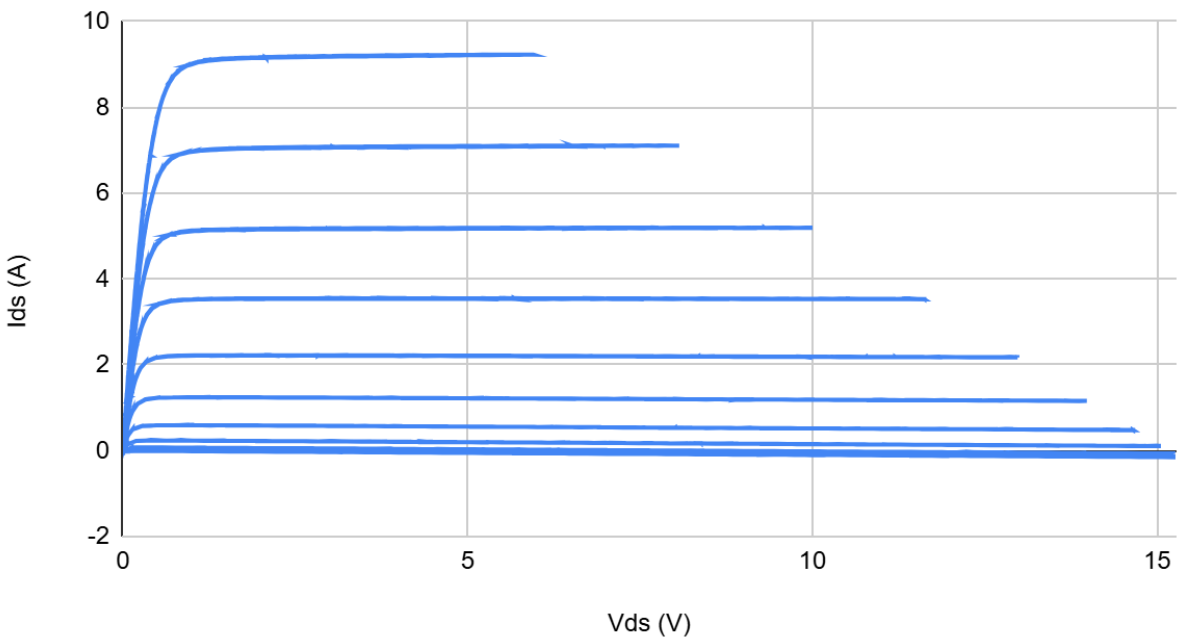


Figure 2.1: Curve Traces graph of n-channel

Curve Trace (p-channel)

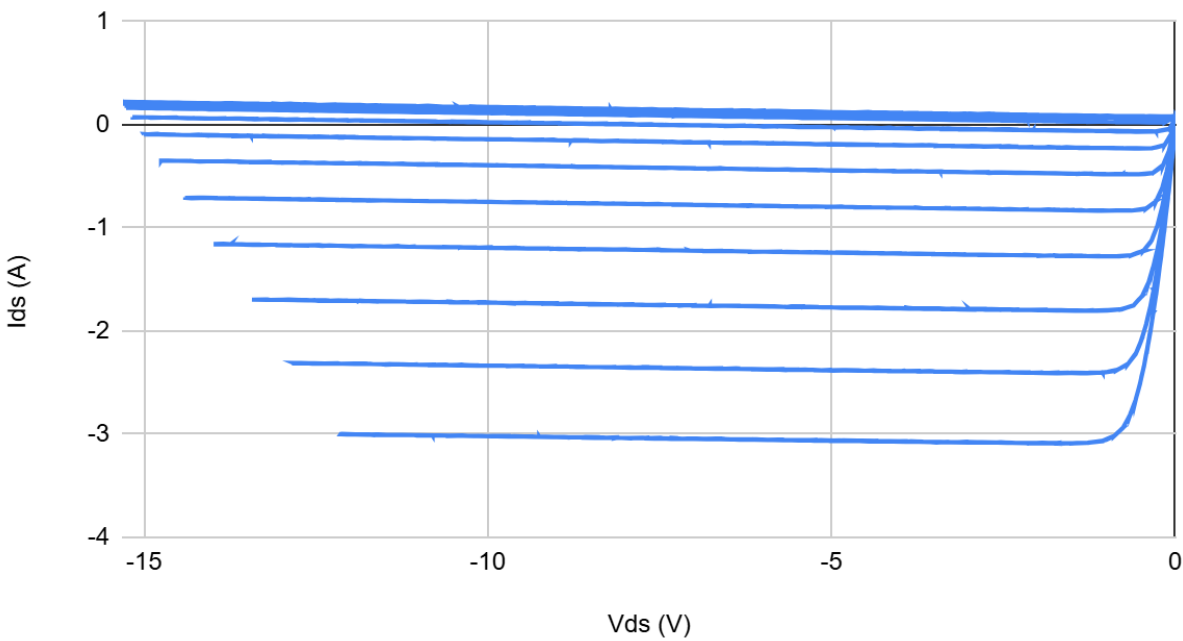


Figure 2.2: Curve Traces graph of p-channel

Part 3 – CMOS Inverter

"CMOS" Inverter Input vs Output characteristics

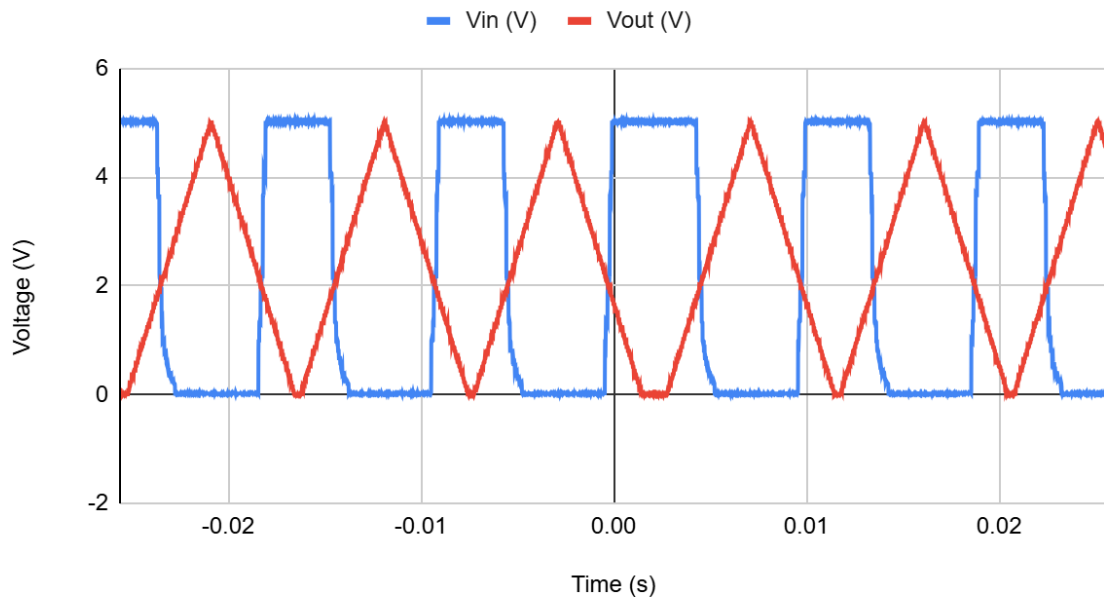


Figure 3.1: CMOS Inverter Input vs Output Characteristics

Voltage Transfer Curve

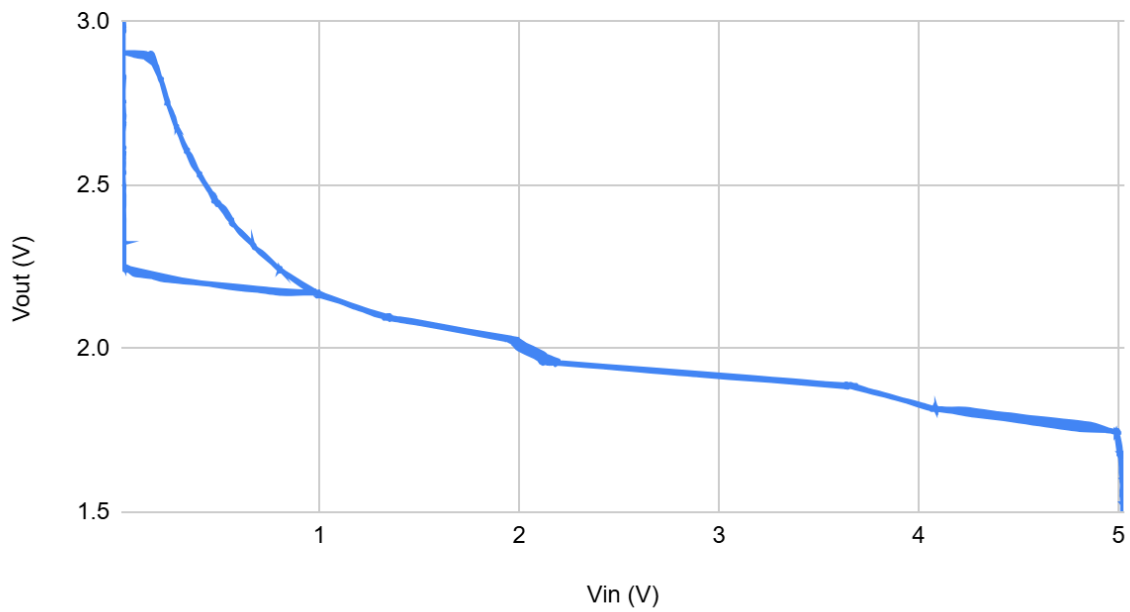


Figure 3.2: CMOS Inverter Voltage Transfer Curve

Part 4 – Nand Gate

Input 2	Input 1	Output
0	0	1
0	1	1
1	0	1
1	1	0

Figure 4.1: NAND Gate Truth Table Results from Circuit

Part 5 – SR Latch

S	R	Q	Q	State
1	0	0	1	Set
0	1	1	0	Reset
1	1	Hold	Hold	Hold
0	0	1	1	Invalid

Figure 5.1: SR Latch Truth Table from Circuit

II. Discussion and Conclusion

This lab provided a hands-on understanding of how MOSFETs operate and how they are used in digital circuits such as inverters, NAND gates, and latches. By experimenting with both n-channel and p-channel MOSFETs, we were able to see how small differences in threshold voltage and device characteristics affect their overall behavior. The threshold voltage tests showed that the n-channel MOSFET began to conduct at around 1.2 volts, while the p-channel device turned on near -2.2 volts. These values were very close to what was expected from the datasheet, confirming the accuracy of our setup and measurements. The moment the gate voltage exceeded the threshold, a clear rise in drain current was observed, marking the transition from the cutoff region to active conduction. Slight variations in reading likely came from measurement tolerances, temperature effects, or small inconsistencies between devices, but overall, the results were consistent and realistic.

The curve tracing part of the experiment further illustrated the three main regions of MOSFET operation: cutoff, triode, and saturation. In the cutoff region, almost no current flowed because the gate voltage was too low to form a conductive channel. As the gate voltage increased, the current began to rise, showing the transition into the triode region where the MOSFET acted more like a voltage-controlled resistor. At even higher voltages, the current leveled off, showing the saturation region where the transistor behaved like a current source. These results lined up very closely with theoretical expectations, and they gave clear visual confirmation of how MOSFETs control current flow using voltage at the gate. This part of the lab really helped connect the theoretical equations to what happens in a circuit.

In the CMOS inverter section, we combined both types of MOSFETs into a single circuit and observed how they worked together to produce an inverted output. When the input voltage was low, the p-channel transistor conducted, pulling the output high. When the input was high, the n-channel transistor conducted instead, pulling the output low. The transition between high and low output was very sharp, occurring around the middle of the supply voltage, which matched the ideal switching behavior we expected. The output also reached nearly the full logic levels of 0 volts and 5 volts, meaning the inverter was efficient and experienced minimal voltage loss. This experiment demonstrated why CMOS inverters are so widely used—they consume almost no power when idle and have strong noise immunity, making them ideal for digital logic applications.

The CMOS NAND gate we built also performed exactly as predicted by its truth table. The output remained high for all input combinations except when both inputs were high, at which point the output dropped to low. The voltage levels were consistent with proper logic operation, and any small rounding at transitions was likely caused by brief switching delays as the transistors changed states. This part of the lab showed how simple logic gates can be built from MOSFETs and how CMOS logic maintains efficiency and reliability while minimizing power consumption.

The SR latch portion tied everything together by showing how logic gates can be combined to create memory. The latch correctly displayed the set, reset, and hold conditions, while the invalid state occurred only when both inputs were low, just as expected. Seeing the circuit “remember” its previous state was a good demonstration of feedback in digital design. It showed that by using basic gates, we can build circuits that not only process signals but also store information, which is a key concept in sequential logic and digital memory systems.

Overall, this lab provided a full picture of how MOSFETs operate individually and within larger CMOS circuits. From observing the threshold voltage and current behavior to building functional digital logic circuits, the experiments confirmed both theoretical and practical aspects of transistor operation. The results showed how nMOS and pMOS devices complement each other in CMOS designs to achieve fast, efficient, and reliable logic switching. More importantly, this lab helped connect circuit theory to real-world applications, demonstrating how the same basic transistor principles form the foundation of modern integrated circuits and computing devices.

III. References

Device Datasheets: ZVN2110A (nMOS), ZVP2110A (pMOS).